

1 Question 1

Done. See code.

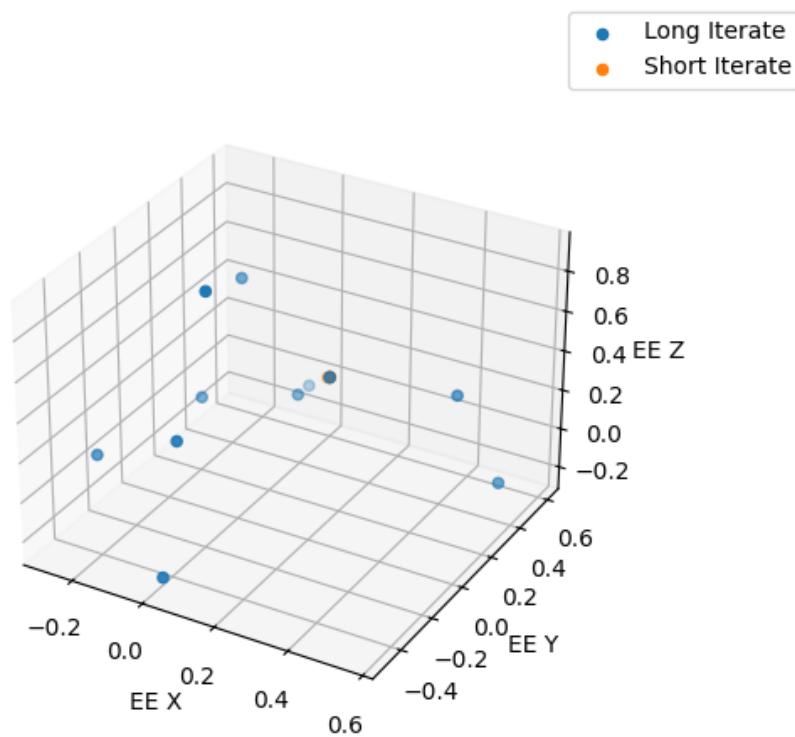


Figure 1: XYZ

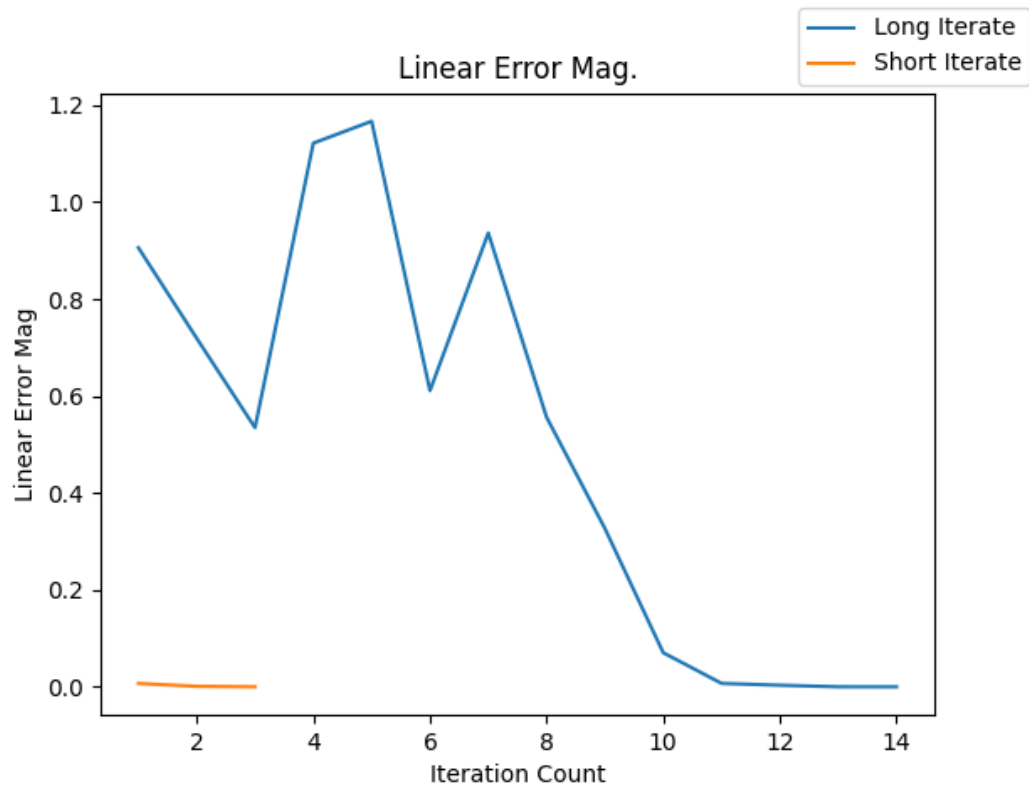


Figure 2: Linear Error Mag

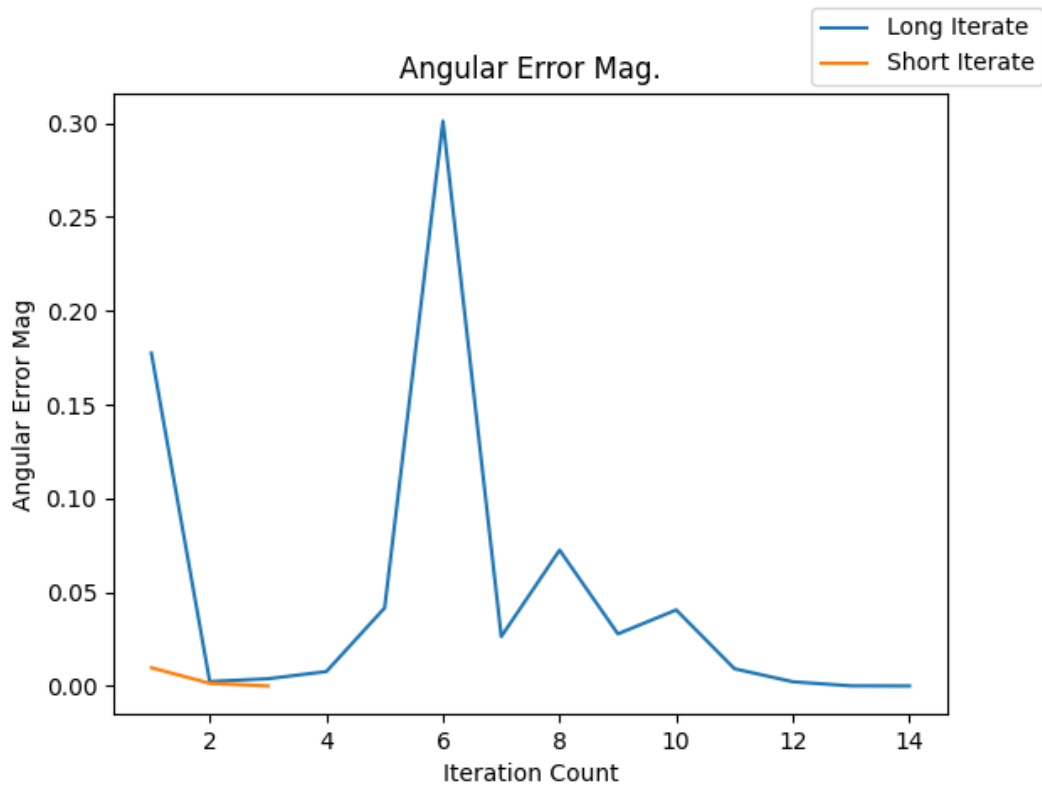


Figure 3: Angular Error Mag

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Iteration: 1
Joint Vector: [[-1.81169434]
[ 2.47631169]
[ 1.32909428]
[ 1.69715231]
[-2.38930106]
[ 2.94686064]]
Tsb: [[ 0.80494566 -0.01236486 -0.59321968  0.19469208]
[ 0.46653451 -0.60455459  0.64564642  0.58656831]
[-0.36661701 -0.79646774 -0.48086496 -0.03874894]
[ 0.          0.          0.          1.          ]]
Error twist: [[ 0.29248522]
[-0.90560653]
[-0.6063037 ]
[-0.08989694]
[-0.06099744]
[ 0.17807711]]
Error rot mag: 1.1283948631496494
Error lin mag: 0.20859914874178317
Iteration: 2
Joint Vector: [[-1.55938949]
[ 2.52574084]
[ 1.13932298]
[ 2.19070709]
[ 3.12805276]
[ 2.71977083]]
Tsb: [[-0.0237574 -0.00547883 -0.99970274  0.01963668]
[ 0.9997023 -0.00568957 -0.02372621  0.64630304]
[-0.00555789 -0.99996881  0.00561237 -0.04653994]
[ 0.          0.          0.          1.          ]]
Error twist: [[-0.00554618]
[ 0.02374429]
[-0.00562431]
[-0.04619792]
[-0.04635474]
[ 0.02031488]]
Error rot mag: 0.025023671592212162
Error lin mag: 0.06852521015238423
Iteration: 3
Joint Vector: [[-1.52931624]
[ 2.65235735]
[ 0.87560342]
[ 1.06289908]
[-3.10383096]
[ 1.44918262]]
Tsb: [[-4.60437581e-02  3.74345891e-02 -9.98237759e-01  3.47375499e-04]
[ 9.98939424e-01  1.72418398e-03 -4.60114644e-02  6.44141404e-01]
[-1.27471027e-06 -9.99297593e-01 -3.74742748e-02  4.54308739e-02]
[ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[ 0.03747645]

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[ 0.04605467]
[ 0.00086196]
[-0.04413864]
[ 0.04539367]
[ 0.00221506]]
Error rot mag: 0.05938232450965684
Error lin mag: 0.06335386175095098
Iteration: 4
Joint Vector: [[-1.52904014]
[ 2.61566038]
[ 1.10168283]
[ 1.97123782]
[ 3.13653414]
[ 2.54699356]]
Tsb: [[-4.59303551e-02 -2.83071683e-03 -9.98940634e-01 1.44168656e-04]
[ 9.98944644e-01 -1.29399810e-04 -4.59301728e-02 6.43373234e-01]
[ 7.52584712e-07 -9.99995985e-01 2.83367279e-03 1.06089790e-02]
[ 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.00000000e+00]]
Error twist: [[-2.83319535e-03]
[ 4.59464898e-02]
[-6.43463365e-05]
[-4.33620649e-02]
[ 1.06110421e-02]
[ 1.12553519e-03]]
Error rot mag: 0.04603380349908945
Error lin mag: 0.04465567952363115
Iteration: 5
Joint Vector: [[-1.52755147]
[ 2.96395045]
[-0.4426156 ]
[-1.62175273]
[-3.10577007]
[-2.24201264]]
Tsb: [[-2.09667855e-02 -2.80328211e-02 -9.99387090e-01 -8.14190451e-03]
[ 9.99780125e-01 -8.98177568e-04 -2.09498374e-02 8.13920284e-01]
[-3.10344023e-04 -9.99606600e-01 2.80454892e-02 -2.70733819e-01]
[ 0.00000000e+00 0.00000000e+00 0.00000000e+00 1.00000000e+00]]
Error twist: [[-0.02804489]
[ 0.0209626 ]
[-0.00060438]
[-0.21406634]
[-0.2707551 ]
[-0.00210258]]
Error rot mag: 0.0350187288381762
Error lin mag: 0.3451624901425448
Iteration: 6
Joint Vector: [[-1.53365312]
[-1.50722725]
[ 2.69738294]
[-1.10259142]

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[-3.11848845]
[-3.05394279]]
Tsb: [[-1.41272425e-02 -2.01859943e-03 -9.99898168e-01  3.30353128e-02]
 [ 9.99900200e-01 -1.37370660e-04 -1.41269939e-02 -1.61322089e-01]
 [-1.08839930e-04 -9.99997953e-01  2.02033864e-03  5.47278982e-02]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[-2.01953759e-03]
 [ 1.41275978e-02]
 [-1.23109474e-04]
 [ 7.61546019e-01]
 [ 5.47125596e-02]
 [ 2.76016713e-02]]
Error rot mag: 0.01427174506280265
Error lin mag: 0.7640076283811125
Iteration: 7
Joint Vector: [[-1.5957586 ]
 [-2.76901493]
 [-1.70997624]
 [-1.62220824]
 [ 3.1167259 ]
 [-2.95954401]]
Tsb: [[ 5.06528272e-04  4.49993576e-03 -9.99989747e-01  3.95390364e-02]
 [ 9.9999865e-01 -1.20060302e-04  5.05993127e-04  5.00862532e-01]
 [-1.17782134e-04 -9.99989868e-01 -4.49999597e-03 -2.31464121e-01]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[ 4.49998125e-03]
 [-5.06262431e-04]
 [-1.18921625e-04]
 [ 9.91137368e-02]
 [-2.31558606e-01]
 [ 3.90432660e-02]]
Error rot mag: 0.0045299310473910295
Error lin mag: 0.25488683248238425
Iteration: 8
Joint Vector: [[-1.51138953]
 [-2.27893964]
 [-1.87383483]
 [-1.51618186]
 [-3.08316679]
 [-2.52717382]]
Tsb: [[-1.16752575e-02 -3.36394720e-02 -9.99365836e-01 -5.15825420e-03]
 [ 9.99931348e-01 -1.38567945e-03 -1.16352211e-02  5.43844742e-01]
 [-9.93398009e-04 -9.99433072e-01  3.36533408e-02  4.73721021e-03]
 [ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[-0.03365353]
 [ 0.01165771]
 [-0.00118979]
 [ 0.05612719]
 [ 0.00461473]
 [-0.00556456]]

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Error rot mag: 0.03563534351666807
Error lin mag: 0.0565908297467378
Iteration: 9
Joint Vector: [[-1.52645597]
[-2.35164929]
[-1.59948693]
[-2.15538669]
[-3.10535446]
[-2.96467653]]
Tsb: [[-8.66792582e-03 -6.36386156e-03 -9.99942182e-01 9.11147225e-04]
[9.99962365e-01 -4.22478191e-04 -8.66541203e-03 5.89804804e-01]
[-3.67308282e-04 -9.99979661e-01 6.36728406e-03 1.40867440e-02]
[0.00000000e+00 0.00000000e+00 0.00000000e+00 1.00000000e+00]]
Error twist: [[-0.0063657 ]
[0.00866684]
[-0.0003949 ]
[0.0102018 ]
[0.01408754]
[0.00082212]]
Error rot mag: 0.010760672717720417
Error lin mag: 0.017412961142769277
Iteration: 10
Joint Vector: [[-1.52548442]
[-2.41715925]
[-1.49809452]
[-2.41103801]
[-3.0969868 ]
[3.0984831 ]]
Tsb: [[-7.47520595e-04 1.92153346e-03 -9.99997874e-01 -4.48553260e-05]
[9.9999720e-01 4.73347910e-05 -7.47431018e-04 5.98863794e-01]
[4.58984767e-05 -9.99998153e-01 -1.92156830e-03 1.66945941e-03]
[0.00000000e+00 0.00000000e+00 0.00000000e+00 1.00000000e+00]]
Error twist: [[1.92155224e-03]
[7.47476336e-04]
[4.66166669e-05]
[1.13615068e-03]
[1.66952861e-03]
[-4.36759636e-05]]
Error rot mag: 0.002062342601633053
Error lin mag: 0.002019918750201
Iteration: 11
Joint Vector: [[-1.52564424]
[-2.41472768]
[-1.50916415]
[-2.3587791 ]
[-3.0964825 ]
[-3.14107843]]
Tsb: [[-4.19383887e-05 -2.31957037e-05 -9.99999999e-01 1.92334985e-06]
[9.99999999e-01 -3.70590660e-07 -4.19383801e-05 5.99987839e-01]
[-3.69617869e-07 -1.00000000e+00 2.31957193e-05 1.01726971e-04]]

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[ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[-2.31957115e-05]
[ 4.19383844e-05]
[-3.70104265e-07]
[ 1.21615014e-05]
[ 1.01726991e-04]
[ 1.92191502e-06]]
Error rot mag: 4.792709150410401e-05
Error lin mag: 0.00010246939342853622
Iteration: 12
Joint Vector: [[-1.52564168]
[-2.41498956]
[-1.50896208]
[-2.35923418]
[-3.09643801]
[ 3.14159215]]
Tsb: [[-5.96976890e-09  2.28952765e-08 -1.00000000e+00 -8.48514536e-10]
[ 1.00000000e+00  5.46857959e-10 -5.96976888e-09  5.99999988e-01]
[ 5.46857723e-10 -1.00000000e+00 -2.28952765e-08  1.54638907e-09]
[ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[ 2.28952765e-08]
[ 5.96976889e-09]
[ 5.46857841e-10]
[ 1.19324460e-08]
[ 1.54638909e-09]
[-8.48514554e-10]]
Error rot mag: 2.3667084352874855e-08
Error lin mag: 1.2062112771523052e-08
Iteration: 1
Joint Vector: [[-1.51203959]
[-2.49402807]
[-1.38000659]
[-2.71707099]
[-3.08946505]
[ 2.8313794 ]]
Tsb: [[-0.00911903  0.01576647 -0.99983412 -0.00850443]
[ 0.99995482  0.00282849 -0.00907553  0.60625615]
[ 0.00268494 -0.9998717 -0.01579155 -0.00857265]
[ 0. 0. 0. 1. ]]
Error twist: [[ 0.0157799 ]
[ 0.00909779]
[ 0.00275687]
[-0.0062831 ]
[-0.00851408]
[-0.00854342]]
Error rot mag: 0.01842214228613973
Error lin mag: 0.01359988740904942
Iteration: 2
Joint Vector: [[-1.52545 ]
[-2.41686269]

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[-1.49568511]
[-2.41475785]
[-3.09839818]
[ 3.09736618]]
Tsb: [[-2.19411368e-03  1.90421659e-03 -9.99995780e-01 -9.07048306e-05]
[ 9.99997582e-01  1.51413570e-04 -2.19382930e-03  5.99307841e-01]
[ 1.47235405e-04 -9.99998176e-01 -1.90454420e-03  2.52924115e-03]
[ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[ 1.90438308e-03]
[ 2.19397458e-03]
[ 1.49324698e-04]
[ 6.91871422e-04]
[ 2.52937867e-03]
[-8.90556491e-05]]
Error rot mag: 0.0029090371716944535
Error lin mag: 0.002623808951810953
Iteration: 3
Joint Vector: [[-1.52564522]
[-2.41444998]
[-1.5096919 ]
[-2.35705745]
[-3.09648558]
[-3.13960686]]
Tsb: [[-4.41192489e-05 -8.95511322e-05 -9.99999995e-01  2.85744527e-06]
[ 9.99999999e-01 -1.83550886e-06 -4.41190847e-05  5.99979845e-01]
[-1.83155794e-06 -9.99999996e-01  8.95512131e-05  1.26007297e-04]
[ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[-8.95511728e-05]
[ 4.41191669e-05]
[-1.83353340e-06]
[ 2.01552257e-05]
[ 1.26007406e-04]
[ 2.85135860e-06]]
Error rot mag: 9.984625818249256e-05
Error lin mag: 0.00012764101913014027
Iteration: 4
Joint Vector: [[-1.52564167]
[-2.41498949]
[-1.50896168]
[-2.35923622]
[-3.09643809]
[ 3.14159056]]
Tsb: [[-8.90087993e-08  9.44737472e-08 -1.00000000e+00 -3.41063833e-09]
[ 1.00000000e+00  2.29236509e-09 -8.90087991e-08  5.99999941e-01]
[ 2.29235667e-09 -1.00000000e+00 -9.44737474e-08  1.44023692e-07]
[ 0.00000000e+00  0.00000000e+00  0.00000000e+00  1.00000000e+00]]
Error twist: [[ 9.44737473e-08]
[ 8.90087992e-08]
[ 2.29236088e-09]
[ 5.89443645e-08]

```



```

[ 1.44023692e-07]
[-3.41063415e-09]]
Error rot mag: 1.2981952928385542e-07
Error lin mag: 1.5565633408876788e-07

}

```

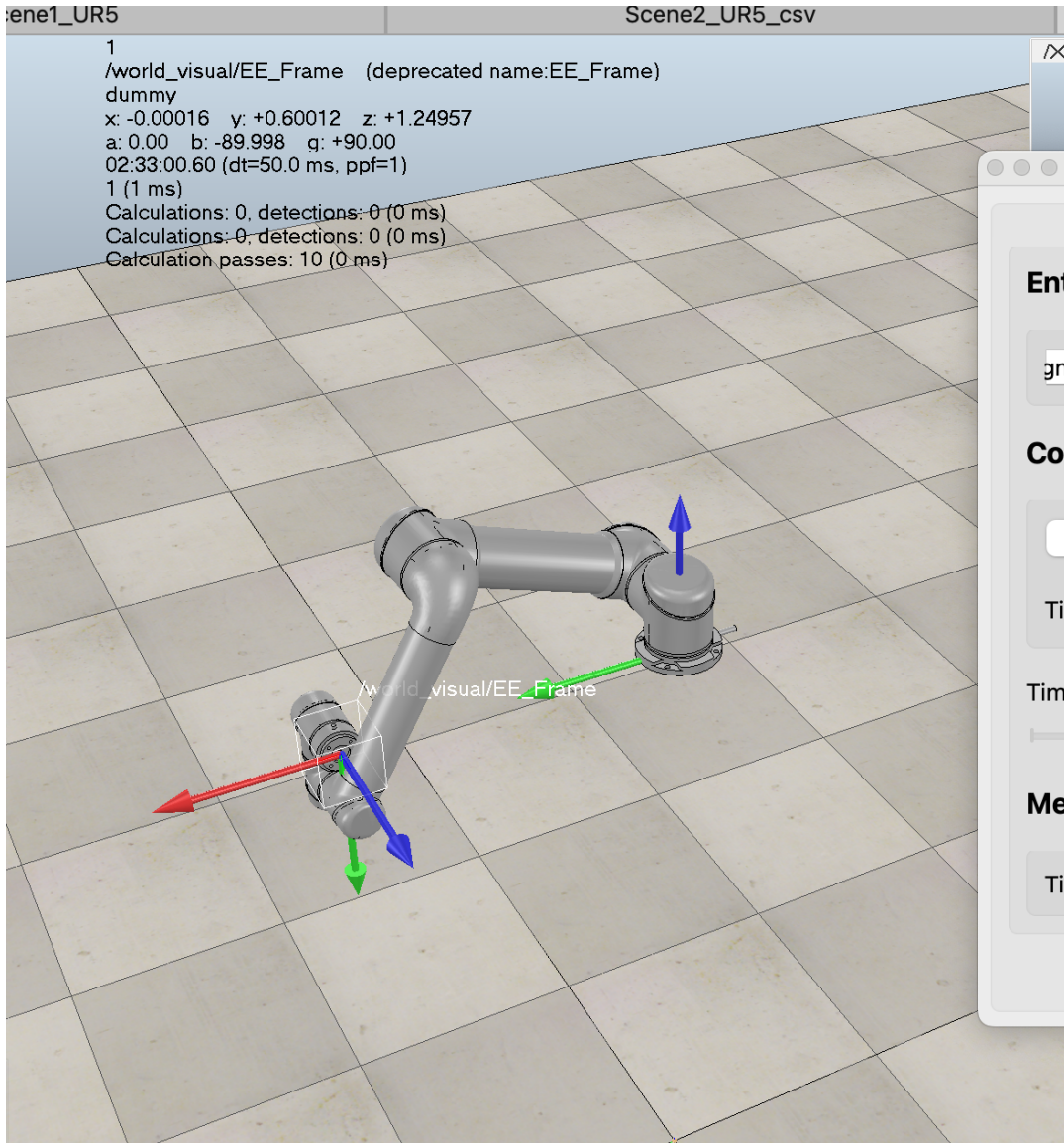


Figure 4: XYZ

2 Part 1 #7

An explanation why convergence is difficult from the long iterates initial guess.

Finding the joint configuration given an end effector configuration is a non-linear process, and is solved numerically via the Newton-Raphson root-finding method. This involves computing locally linearized gradients and then

using those gradients to estimate the zero value of the higher order function of interest.

Since the higher order function is non-linear, the first-order gradient may not be consistent. This may cause erratic gradient values when the input variable is far from a local minima.

3 Part 2

a) I'd compute the joint states for the initial transform, and the joint states for the final transform, and then linearly interpolate between the two joint state vectors

$$\dot{\theta} = (\theta^* - \theta^0)/t_f \quad (1)$$

b) I'd compute the unit twist between transforms, scale it to t_f , then use that to compute the average joint speed I'm assuming the FK and IK equations are given.

Forward Kinematics:

$$T_{sd}(\theta) = FK(\theta^*) \quad (2)$$

$$T_{s0}(\theta) = FK(\theta^0) \quad (3)$$

Finding the desired transform

$$T_{0d}(\theta) = T_{s0}^{-1} \cdot T_{sd} \quad (4)$$

Find the desired twist

$$[\mathcal{V}] = \log(T_{0d}) \quad (5)$$

In the body frame

$$\mathcal{V} = \text{vec}([\mathcal{V}]) \quad (6)$$

Scale it to t_f

$$\mathcal{V}_d = \mathcal{V}/t_f \quad (7)$$

Eq 6.7 in M.R.

$$\dot{\theta}^0 = J_B^\dagger(\theta^0) \cdot \mathcal{V}_d \quad (8)$$

For finding the angular velocity halfway through.

Get the transform halfway through. Left multiply because the desired twist is expressed in the body frame

$$T_{1/2} = \exp([\mathcal{V}_d] * 1/2 * t_f) * T_{0d} \quad (9)$$

Do IK to get the joint state halfway through

$$\theta^{1/2} = IK(T_{1/2}) \quad (10)$$

Use the Jacobian and the desired twist to get the angular velocities

$$\dot{\theta}(t = 1/2 \cdot t_f) = J_B^\dagger(\theta^{1/2}) \cdot \mathcal{V}_d \quad (11)$$

c) Yes, option A is conceptually easier to understand and implement, and therefore less bug prone. You can also use the current config as given, so only one IK problem must be solved (the goal config).

However, option B ensures the end effector will move in a straight line from the initial config to the goal, meaning that its motion will be predictable in euclidean space.

This is especially pertinent if there are obstacles and/or objects in the environment that the robot arm wishes to avoid for safety reasons.